

# Mustafa Alhelawe

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## SUMMARY

Full-stack software engineer with 3+ years of experience building high-volume financial systems. One of two U.S.-based engineers responsible for a global institutional trading platform processing millions of trades daily, with additional experience building independent software projects across multiple domains.

## SKILLS

**Languages:** Scala, JavaScript, TypeScript, Python, C#, Java, Bash, SQL

**Frameworks & Backend:** React, Node.js, Next.js, REST APIs, PostgreSQL

**AI / ML:** PyTorch, Scikit-learn, Reinforcement Learning, LLMs

**Tools:** Git, Docker, Linux, CI/CD, Jenkins, Gradle, Cypress, Splunk, Unity, Claude, Codex

## WORK EXPERIENCE

### Software Engineer

**Jersey City, New Jersey**

*Bank of America Merrill Lynch*

January 2023 – Present

- Developed and maintained allocation workflows supporting 10+ trade flows across matching, booking, confirmation, amendment, and cancellation, including Mexico's onboarding onto the platform under the Mexico T+1 initiative.
- Built services to ingest and link orders from multiple upstream systems through FIX-based messaging and file-based workflows, supporting real-time trade matching and confirmation.
- Drove initial development of self-service onboarding, building configuration APIs that let operations teams onboard clients and amend settings immediately instead of waiting for biweekly production releases.
- Migrated 300+ client configurations from release-managed YAML files to a new JSON schema in AMPS SOW, providing the configuration data model used by the self-service onboarding APIs.
- Designed configuration approval workflows with current-versus-proposed diffs and self-approval safeguards.
- Implemented bulk approval and rejection workflows with partial-failure handling, allowing operations users to process multiple work items together instead of approving or rejecting each individually.
- Led a zero-downtime rollout with global engineering teams, doubling the production footprint to 15+ servers.

### Cybersecurity Analyst Intern

**East Rutherford, New Jersey**

*Capri Holdings Limited*

June 2022 – August 2022

- Conducted vendor risk assessments, reviewing corporate partners' security policies for protecting confidential information.
- Investigated and triaged LogRhythm security alerts, including unauthorized communications with banned IP addresses.
- Analyzed suspicious inbound emails for indicators of compromise using CrowdStrike's sandbox.

## PROJECTS

### *Splits — Automated Shared Expense Tracking*

February 2026 – Present

- Built an expense-sharing application with a Next.js web app and React Native/Expo mobile clients, supporting recurring bills, customizable expense splits, and balance tracking.
- Consolidated three implementations of expense-calculation logic into one shared TypeScript package, with fixture-based tests for split calculations, rounding, balances, and debt simplification.
- Detected recurring bills from Plaid transactions using normalized merchant names, amount clustering within a 15% tolerance, and recurrence across 2+ months; generated suggested split rules for automatic shared-bill entries.
- Built incremental Plaid transaction syncing with per-item concurrency locks and recovery from pagination errors, restarting paginated retrieval when transaction data changed during a sync.
- Applied generated bill entries before advancing the transaction-sync cursor, using database-enforced idempotency keys to prevent duplicate ledger entries when a sync is retried or replayed.
- Moved the mobile backend into Next.js API routes so web and React Native/Expo clients share Supabase authentication and PostgreSQL storage, eliminating a separate backend deployment.

### *Reinforcement Learning Simulations*

January 2025 – Present

- Built a series of reinforcement-learning simulations in Unity ML-Agents, implementing modular C# environment logic and Python training pipelines to train autonomous agents using PPO.
- Progressed from food-seeking without vision to navigation around static and moving obstacles, then visual observations, refining observation design and reward functions through successive training experiments.
- Extended reinforcement-learning experiments to a 6-axis robotic arm, applying the same simulation approach to more complex observations, action spaces, reward functions, and robotic control.

## EDUCATION

**Rutgers, The State University of New Jersey**

**New Brunswick, New Jersey**

*Bachelor of Science in Electrical and Computer Engineering*

Class of 2022